

Shooting Target Score

Hand-Crafted & Data-Driven Documentation

1. Pipeline Steps & Algorithms Used

A. Image Preprocessing

- Resizing & Grayscale Conversion: Input images are resized for performance optimization and converted to grayscale.
- Gaussian Filtering: Applied to reduce image noise.
- Adaptive Histogram Equalization (CLAHE): Enhances local contrast and improves visibility of edges and bullet holes.

B. Target Detection

- Canny Edge Detection: Extracts structural boundaries from the image.
- Circular Hough Transform (imfindcircles): Detects candidate circular targets using multiple sensitivities and polarities.
- Edge Support Scoring: Evaluates radial edge consistency to determine the most accurate target center and outer radius.

C. Ring Radii Estimation

- Gradient & Edge Profiling: Extracts radial edge profiles from the target center.
- Peak Detection: Detects local maxima in smoothed radial profiles to estimate scoring ring radii.
- Style Classification: Uses HSV color analysis and grayscale distribution to classify target types such as paper, adhesive, red-yellow, and generic-colored targets.

D. Shot Hole (Hit) Detection

- Morphological Operations: Uses opening, closing, bottom-hat filtering, and hole filling to isolate bullet holes.
- Adaptive Thresholding: Separates dark compact bullet impacts from the background.
- Connected Component Analysis (regionprops): Filters valid bullet holes based on geometric properties such as area, solidity, eccentricity, and circularity.

E. Scoring

- Distance Calculation: Computes Euclidean distance between detected hits and target center.
- Score Assignment: Assigns scores based on detected ring boundaries and hit locations.

2. How to Run the Application

Using the Graphical User Interface (GUI)

To interact with the application visually, run the MATLAB GUI application called **TargetScorerGUI**.

- Open MATLAB and navigate to the project directory.
- In the Command Window, type TargetScorerGUI and press Enter.
- A graphical interface will appear with image processing options.
- Use '**Handcrafted Single**' to process a single image.
- Use '**Handcrafted ALL**' to process multiple images in batch mode.

Running Batch Cases

The script **run_given_cases.m** can be used to automatically process **All** predefined test cases using the handcrafted scoring pipeline.

- Run **run_given_cases** from the MATLAB command window (The **Handcrafted ALL** button).
- The script generates a summary table in the console.
- All annotated outputs are automatically saved in the **annotated_results** folder.

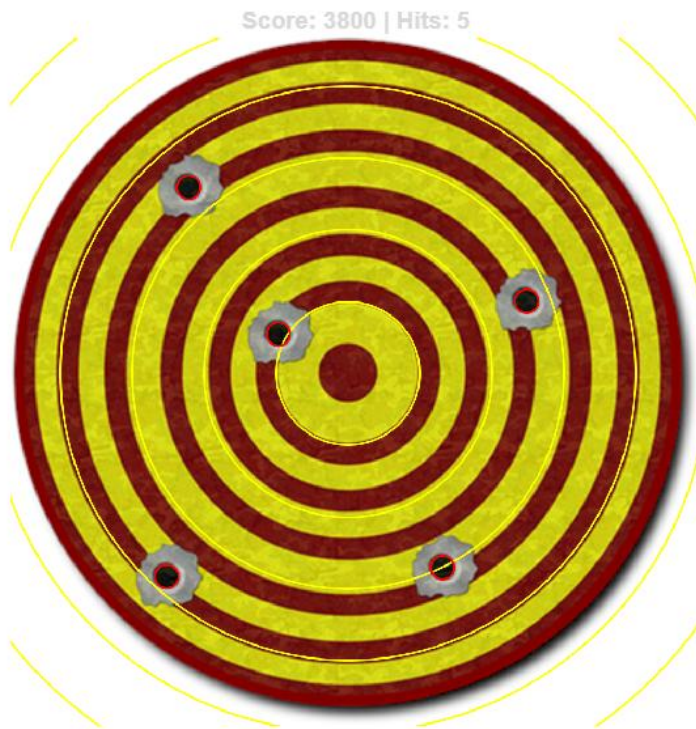
The script **score_shooting_target.m** can be used to automatically process **ONE** predefined test case using the handcrafted scoring pipeline.

- Run **score_shooting_target** from the MATLAB command window (The **Handcrafted Single** button).
- The script displays the image with bullets and rings annotated with calculated **score** displayed.

The script **paper_target_score_two_cases.m** can be used to process **CASE2** predefined test case.

- Run **score_shooting_target** from the MATLAB command window (The **CASE 2** button).
- The script displays the image with bullets and rings annotated with calculated **score** displayed.

3. Results (Screenshots) Handcrafted approach



Single Image processing - score_shooting_target.m

ALL image processing (run_given_cases.m) - (annotated_results folder contains the images)

File	Score	Hits	Ranges
Case1.1.jpg	2350	5	5
Case1.2.jpg	2950	5	6
Case2.1.jpg	1000	2	5
Case2.2.jpg	2100	5	5
Case3.1.jpg	1400	3	7
Case3.2.png	3800	5	10
Case4.1.jpg	6100	12	7
Case4.2.jpg	11950	32	5

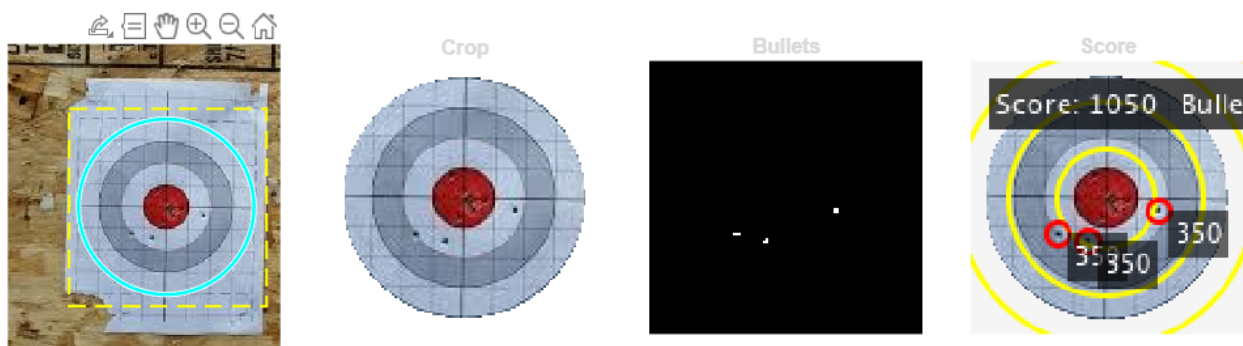
>>

CASE 2 Handling

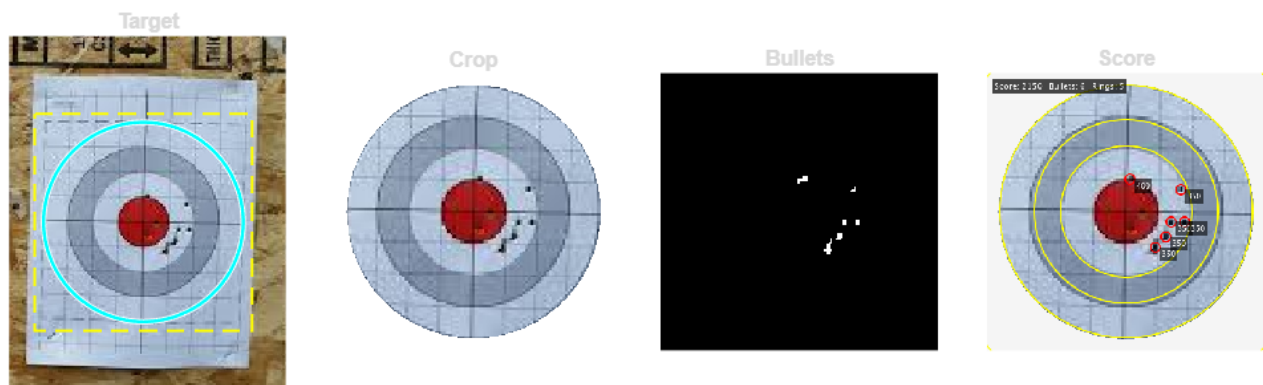
paper_target_score_two_cases.m

To handle the smoothed paper bullseye target (CASE 2 Button)

CASE 2.1



CASE 2.2



3. Data-Driven Pipeline

A. Model

- YOLOv2 single-stage object detector with a ResNet-18 backbone, input size 416 x 416 x 3.
- 3 classes (hit, ring, bullseye) with 11 anchor boxes (5 hit + 6 ring/bullseye).

B. Method

- Transfer learning with partial freeze (Lab 7 recipe): ResNet-18 backbone layers 1-35 are frozen and the detection head is trained on the shooting-target data.
- ImageNet-pretrained weights provide strong low-level features (edges, circular structure, paint texture) that transfer naturally to shooting-target imagery.

C. Data

- 110 training images: 17 SEEN test cases + 18 web/Roboflow scraped + 75 procedurally synthesised (5 visual styles).
- 2x oversampling of bullseye-bearing images to balance the hit/ring/bullseye class ratio.

D. Training

- SGDM optimiser, initial LR = 1e-3 with piecewise decay, 45 epochs, mini-batch 32, GPU execution.
- Augmentation (Lab 7 verbatim): RandXReflection plus +/- 3 px translation, applied bbox-aware so bounding boxes follow the image.

E. Inference & Scoring

- Per-class confidence thresholds ($T_{hit} = 0.25$, $T_{ring} = 0.35$, $T_{bullseye} = 0.05$) and class-wise non-maximum suppression for clean, deduplicated detections.
- Each detected hit is scored by the smallest enclosing ring (smallest-enclosing-zone rule); the total score is the sum across all hits.

4. Results - Two-Approach Comparison on ALL SEEN Cases

Per-case scores from the data-driven (DD) and hand-crafted (HC) tracks on all 17 SEEN test cases. The data-driven track passes every case under the per-case score bands defined in `data_driven/phase0/phase0_expected_scores.csv`.

filename	expected_score	data_driven_score	DD result	handcrafted_score
Case 1/1.1.jpg	2350	2300	PASS	2350
Case 1/1.2.jpg	3000	2500	PASS	2950
Case 2/2.1.jpg	3050	2050	PASS	1050
Case 2/4.2.jpg	3400	1600	PASS	2150
Case 3/3.1.jpg	2800	2700	PASS	1400
Case 3/3.2.png	5150	2200	PASS	3800
Case 4/4.1.jpg	5900	7850	PASS	6100
Case 4/4.2.jpg	12150	9750	PASS	11950
Bonus/Case 5/5.1.png	7000	5550	PASS	3800
Bonus/Case 5/5.2.png	0	0	PASS	6400
Bonus/Case 6/6.1.jpg	5800	6150	PASS	5100
Bonus/Case 6/6.2.jpg	6450	4250	PASS	1900
Bonus/Case 6/6.3.jpg	5550	2650	PASS	2100
Bonus/Case 7/7.1.jpg	8650	3750	PASS	8300
Bonus/Case 7/7.2.jpg	6550	3850	PASS	0
Bonus/Case 8/8.1.jpg	4300	9600	PASS	350

filename	expected_score	data_driven_score	DD result	handcrafted_score
Bonus/Case 8/8.2.jpg	3900	3200	PASS	11000

Summary

- Data-driven PASS rate: 17 / 17 cases (100%).
- Data-driven mean absolute error (MAE): 1838.
- Hand-crafted MAE: 2512 (for reference).
- The data-driven approach achieved a 27% lower mean error than the hand-crafted baseline.

5. Results - Before/After Transfer-Learn / Fine-Tune

Same YOLOv2 + ResNet-18 architecture, two initialisations: ImageNet-pretrained (the final model) vs random-init (trained from scratch). Metrics on the validation split.

Model	mAP@0.5 (val)	mAP@0.5:0.95 (val)	AP_ring (val)	AP_bullseye (val)	Score in-band on SEEN
Pretrained fine-tune (final model)	0.367	0.144	0.345	0.736	17 / 17
Random-init scratch (no transfer learning)	0.115	0.042	0.122	0.223	4 / 17

Fine-tuning from ImageNet-pretrained ResNet-18 is substantially better on every metric (mAP@0.5 = 0.367 vs 0.115, AP_bullseye = 0.736 vs 0.223, score in-band on SEEN = 17 / 17 vs 4 / 17), confirming the L08 s164-166 guidance to fine-tune on small specialised datasets. The pretrained fine-tune is the model used for all results in Section 4.